



THE UNIVERSITY OF BRITISH COLUMBIA

College of Graduate Studies

Belief Distribution Learning - an extension to Evidential Deep Learning

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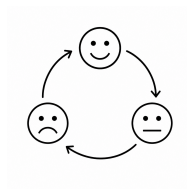


What do these have in common?

States of a Hidden Markov Model



Facial Expressions

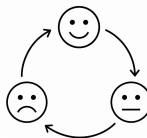


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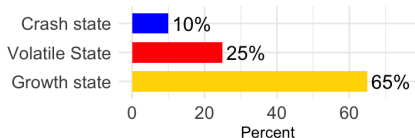


Facial Expressions

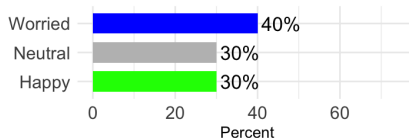


They both represent Label Distribution Learning

State Forecast

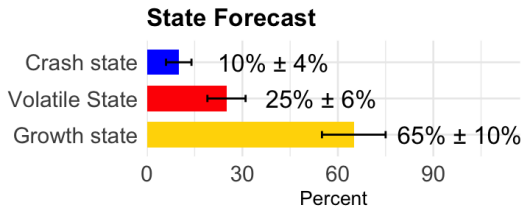


Facial Expression Distribution



Develop an uncertainty aware Deep Learning classifier for LDL applications.

- ▶ Prioritize accuracy
- ▶ Provide calibrated uncertainty estimation



Generic Prediction vs Evidential Deep Learning

Softmax prediction output:

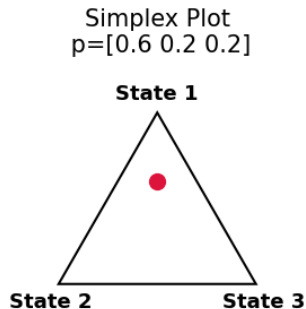


Figure: Point estimate with no uncertainty prediction.

Dirichlet distribution output:

Dirichlet Distributions with Increasing Evidence

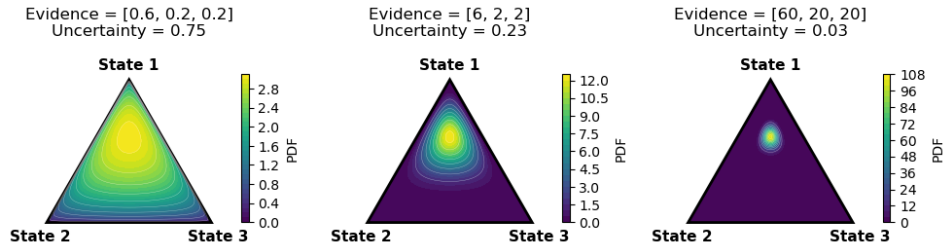


Figure: Dirichlet distribution with uncertainty quantification.

Evidential Deep Learning

In Evidential Deep Learning [Sensoy 2018], a neural network with a ReLU output layer learns an evidence vector $\mathbf{e}$ that parametrizes a Dirichlet Distribution.

Step 1: The network maps inputs to evidence:

$$\mathbf{e} = f_{\hat{\theta}}(X) \quad (\text{ReLU ensures } e_i \geq 0) \quad (1)$$

Step 2: Evidence defines the Dirichlet parameters:

$$\alpha_i = e_i + 1 \quad \forall i \in K \quad (2)$$

$$\text{Dir}(\mathbf{p}, \alpha) = \frac{1}{B(\alpha)} \prod_{i=1}^K p_i^{\alpha_i - 1} \quad (3)$$

$$\text{Model Uncertainty : } u = \frac{K}{\sum_{i=1}^K \alpha_i} \quad (4)$$

Evidential Deep Learning

The Problem: Learned evidence vectors treat each class independently and ignore the *simplex constraint*.

Because probabilities must sum to one, evidence towards one class *requires* reduced evidence towards the other classes.

Example: If the model is certain that $p_1 = 1$, then it must be equally certain that $p_2 = 0$, but standard EDL does not enforce this.

Belief Distribution Learning

Belief Distribution Learning

Belief Distribution Learning is derived from *Subjective Logic* [Jøsang 2016], which decomposes a probability into belief and uncertainty:

$$\mathbf{P} = \mathbf{b} + u \mathbf{a} \quad (5)$$

$$1 = u + \sum \mathbf{b} \quad (6)$$

where

- ▶ $\mathbf{b}$ — the *belief* mass assigned to each outcome
- ▶ u — the *uncertainty* mass
- ▶ $\mathbf{a}$ — the *base rate* (prior probability)

The probability of an event is your belief, plus your uncertainty distributed according to the prior.

Belief Distribution Learning

We propose Belief Distribution Learning, where a deep learning model predicts the belief $\mathbf{b}$ and uncertainty u through a **softmax** output layer.

$$u, \mathbf{b} = f_{\hat{\theta}}(X) \quad (\text{Softmax ensures } \sum b_i + u = 1) \quad (7)$$

These outputs map to **Dirichlet parameters**, which provide uncertainty quantification:

$$\alpha_i = \frac{1}{K} \left(\frac{b_i}{u} + \bar{p}_i \right) + 1 \quad (8)$$

where K is the number of classes, and $\bar{p}_i$ is the average class probability over the training data.

Example PDF plots

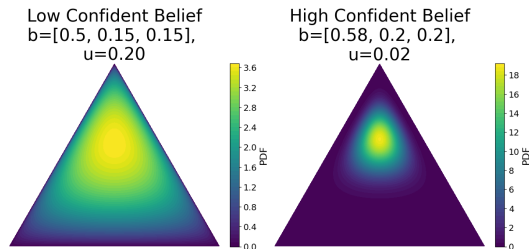


Figure: Sample low confidence and higher confidence predictions using Belief Distribution Learning.

Table: Label distribution prediction error on SJAFPE dataset ($\downarrow$ lower is better)

Model	KL Div. $\downarrow$	Chebyshev $\downarrow$	Clark $\downarrow$	Canberra $\downarrow$
BDL	0.042 ± 0.003	0.087 ± 0.004	0.319 ± 0.009	0.654 ± 0.019
EDL	0.053 ± 0.004	0.103 ± 0.004	0.357 ± 0.010	0.741 ± 0.023
DP	0.073 ± 0.004	0.119 ± 0.005	0.424 ± 0.012	0.886 ± 0.026
SNEFY	0.136 ± 0.011	0.146 ± 0.007	0.566 ± 0.021	1.172 ± 0.044

BDL (ours): Belief Distribution Learning **EDL**: Evidential Deep Learning [Sensoy 2018]

DP: Direct Prediction **SNEFY**: Squared Neural Family [Zhang 2025]

Takeaway: BDL achieves the lowest error across all four metrics.

multi-Belief Distribution Learning

multi-Belief Distribution Learning

The Idea: What if we deconstruct multi label learning into K single label learners.

- ▶ Each learner is a 1 vs rest classifier
- ▶ The learners are constrained to the simplex using belief logic

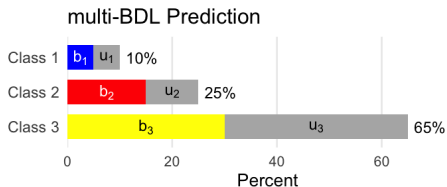
From belief logic we define:

$$p_i = b_i + u_i \quad \forall i \in K \quad (9)$$

$$1 = \sum_{i=1}^K b_i + u_i \quad (10)$$

$$\mathbf{b}, \mathbf{u} = f_{\hat{\theta}}(X) \quad (\text{Softmax}) \quad (11)$$

Multi-Belief Distribution Learning



Construct K Beta distributions (**one-vs-rest**):

- ▶ Class i : $\alpha_i = \frac{1}{K} \left(\frac{b_i + u_i}{u_i} \right) + 1$
- ▶ The rest: $\beta_i = \frac{1}{K} \left(\frac{1 - b_i - u_i}{u_i} \right) + 1$

The joint PDF is the product over labels:

$$\text{mBDL}(\mathbf{p} \mid \boldsymbol{\alpha}, \boldsymbol{\beta}) = \prod_{i=1}^K \text{Beta}(p_i \mid \alpha_i, \beta_i)$$

Multi-Label Belief Distribution Learning

Multi-BDL has flexible confidence bounds while BDL does not.

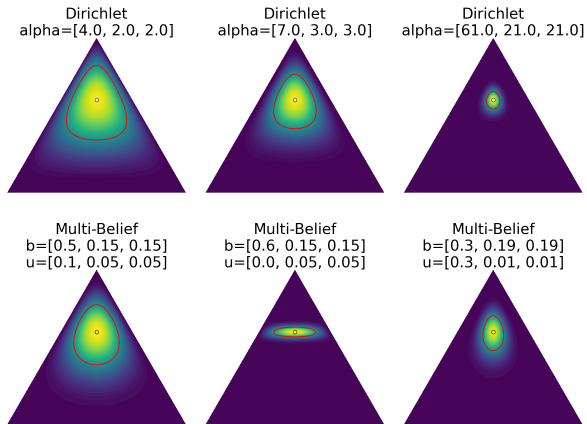


Table: Label distribution error and conformal calibration on SJAFEE

Model	KL Div. ↓	Joint FSC ↑	Worst-Bin FSC ↑
BDL	0.042 ±0.003	0.551±0.034	0.518±0.039
multiBDL	0.042 ±0.002	0.586 ±0.036	0.568 ±0.030
EDL	0.060±0.005	0.510±0.034	0.488±0.022
DP	0.073±0.004	0.498±0.045	0.547±0.031
SNEFY	0.141±0.013	0.571±0.045	0.540±0.026

KL Divergence: Measure of agreement between predicted and true label distribution. **FSC (Feature-Stratified Coverage):** Uncertainty calibration metric.

Joint Combined calibration across labels; **Worst-Bin** Worst performing quartiles across labels

Takeaway: multiBDL is equally accurate but much better calibrated.

Conclusions

- ▶ Belief theory from *Subjective Logic* applied to the concepts of Evidential Deep Learning improves accuracy on LDL tasks.
- ▶ Multi-belief, which allows for label specific uncertainty, improves conformal predictions compared to competing methods.

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SNEFY	0.141±0.013	0.571±0.045	0.540±0.026

Evidential Deep Learning to Quantify Classification Uncertainty

Author: Murat Sensoy, Lance Kaplan, Melih Kandemir **Proceedings:** 32nd Conference on Neural Information Processing Systems. **Year:** 2018

Label Distribution Learning using the Squared Neural Family on the Probability Simplex

Author: Daokun Zhang, Russell Tsuchida, Dino Sejdinovic **Proceedings:** Forty-first Conference on Uncertainty in Artificial Intelligence. **Year:** 2025

Subjective Logic: A Formalism for Reasoning Under Uncertainty

Author: Audun Jøsang **Publisher:** Springer, Artificial Intelligence: Foundations, Theory, and Algorithms series. **Year:** 2016

Table: Accuracy and conformal metrics for SBU_3DFE dataset

model	dataset	kl_divergence	joint_fsc	worst_bin_fsc
EDL	SBU_3DFE	0.067 ± 0.001	0.488 ± 0.012	0.762 ± 0.013
multiBDL	SBU_3DFE	0.067 ± 0.001	0.482 ± 0.013	0.790 ± 0.010
BDL	SBU_3DFE	0.070 ± 0.001	0.482 ± 0.011	0.745 ± 0.013
DP	SBU_3DFE	0.076 ± 0.001	0.480 ± 0.008	0.765 ± 0.007
SNEFY	SBU_3DFE	0.409 ± 0.050	0.575 ± 0.010	0.734 ± 0.018

Uncertainty types

Epistemic Uncertainty: When the model has insufficient information to make a confident prediction.

- Generally dominates in out-of-distribution examples.
 - Example: Covid-19 was a novel environment with high epistemic uncertainty.
- Predictions should have low confidence.

Aleatoric Uncertainty: Randomness inherent to the data itself.

- Should lead to more probability mass spread across the hidden states.
 - Example: Random week-to-week swings of an asset price.
- Predictions can have high confidence.